PingSheng Kao

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SUMMARY

Research-focused ML Data Scientist with 6+ years of experience designing datasets, benchmarks, and evaluation systems for multimodal, LLM, agentic, and production ML systems. Translates ambiguous research questions into expert-curated data, rigorous experiments, scalable evaluators, and decision-ready model insights.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Designed an expert-curated gold-set and labeling workflow for a multimodal health AI product, combining expert verification, representative user feedback, and an auditable multi-stage LLM evaluator.
- Reached 90% evaluator alignment with health experts on a 40-example gold set and deployed the evaluator internally for performance reporting and systematic failure discovery.
- Led cross-functional translation from ambiguous user-research needs into engineering requirements and Data Operations labeling instructions, coordinating LLM-as-judge evaluation with human-in-the-loop validation.
- Built simulated-user, agent-as-judge, targeted-scenario, and behavioral-metric workflows for a multimodal conversational agent; evaluation findings changed subsequent development direction.
- Contributed to large-scale embedding-index experiments that curated an internal segmentation training dataset aligned with Meta's Segment Anything research and delivered it to a foundation-model team.
- Developed embedding-driven error-slice discovery, fine-tuned vision models on targeted failure data, and produced leadership-facing reports connecting data slices, regression attribution, and online validation to model decisions.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments to ECCV-accepted research on grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming & Data: Python, SQL, C++, JavaScript, AWS Athena
Evaluation: LLM-as-Judge, Agent-as-Judge, Benchmark Design, Human Evaluation, DSPy
Datasets & Experiments: Gold Sets, Data Curation, Embedding Indexing, Vision-Model Fine-Tuning
Analysis: Metric Design, Bayesian Analysis, Failure Slices, Clustering, OOD Analysis, Tableau